CPL rotation and pin-1 check table (r7d)

來源：v2_r7f/fab_r7f/CPL_pin1_check.md（原文英文）

r7b M2: one row per placed part: CPL angle, KiCad angle, and where pin 1 must appear in JLC's placement preview.

Pin 1 is read from the saved board (pad "1", or A1 for the BGA, or the first pad of the connectors), as an offset from the part centre in board coordinates seen from the top (x right, y toward the back wall = down in KiCad and in JLC's top preview). For bottom-side parts the same offset is also given as seen in JLC's bottom preview, which shows the board flipped about the vertical axis (x mirrored). The JLC angle convention is per LCSC part (r6/r7 history: C2871228 at 0 and C78988 at 270 on the same SOT-23-5 land), so nothing here predicts JLC's angle; the table only says what the preview must show. "verified" means checked by a person in JLC's preview (2026-11-07).

Ref	LCSC	Land	Side	KiCad deg	CPL deg	CPL-KiCad	Pin	Pin 1 from centre, top view (mm)	Same pin in JLC bottom view	Status
C69	C347552	CP_7343	Bottom	0	0.0	0.0	1	right (+3.10, +0.00)	left (-3.10, +0.00)	CHECK in preview; new in r7c: polarized (pin 1 = +, silk + beside it)
D1	C48260	X1SON-2_0.6x1	Top	0	0.0	0.0	1	left (-0.35, +0.00)		CHECK in preview
J2	C720616	HDMI-001S	Bottom	180	180.0	0.0	1	down-right (+4.50, +3.36)	down-left (-4.50, +3.36)	CHECK in preview
J3	C41427493	USB-C_HX_16P	Top	180	180.0	0.0	A1 B12	down-right (+3.20, +2.55)		CHECK in preview
J4	C160405	JST_SM06B	Bottom	0	0.0	0.0	1	up-right (+2.50, -2.00)	up-left (-2.50, -2.00)	CHECK in preview
J5	C41427493	USB-C_HX_16P	Top	90	90.0	0.0	A1 B12	down-left (-2.55, +3.20)		CHECK in preview
J6	C160407	JST_SM08B	Top	90	90.0	0.0	1	down-left (-2.00, +3.50)		CHECK in preview
J7	C585350	microSD_Molex_104031-0811	Bottom	270	90.0	180.0	1	up-left (-5.45, -3.10)	up-right (+5.45, -3.10)	CHECK in preview (seen right on 2026-11-07; re-check, review M2)
J8	C88373	U.FL-R-SMT-1	Bottom	0	0.0	0.0	1	down (+0.00, +1.52)	down (-0.00, +1.52)	CHECK in preview
Q1	C15127	SOT-23	Bottom	0	270.0	270.0	1	down-right (+0.95, +0.94)	down-left (-0.95, +0.94)	CHECK in preview (seen right on 2026-11-07; re-check, review M2)
U10	C92243	SSOP-4_1.27	Bottom	0	270.0	270.0	1	up-right (+3.17, -0.63)	up-left (-3.17, -0.63)	CHECK in preview (seen right on 2026-11-07; re-check, review M2)

Ref	LCSC	Land	Side	KiCad deg	CPL deg	CPL-KiCad	Pin	Pin 1 from centre, top view (mm)	Same pin in JLC bottom view	Status
U11	C92243	SSOP-4_1.27	Bottom	0	270.0	270.0	1	up-right (+3.17, -0.63)	up-left (-3.17, -0.63)	CHECK in preview (seen right on 2026-11-07; re-check, review M2)
U12	C92243	SSOP-4_1.27	Bottom	0	270.0	270.0	1	up-right (+3.18, -0.64)	up-left (-3.18, -0.64)	CHECK in preview (seen right on 2026-11-07; re-check, review M2); moved 0.08 mm in r7c
U14	C138714	USON-10	Top	0	0.0	0.0	1	down-left (-1.00, +0.41)		CHECK in preview
U15	C78988	SOT-23-5	Top	90	0.0	270.0	1	down-right (+1.14, +0.95)		CHECK in preview
U16	C2155674	WSON-8_2x2_EP	Bottom	90	90.0	0.0	1	down-right (+0.75, +0.95)	down-left (-0.75, +0.95)	CHECK in preview
U2	C78988	SOT-23-5	Bottom	0	270.0	270.0	1	down-right (+0.95, +1.14)	down-left (-0.95, +1.14)	CHECK in preview (seen right on 2026-11-07; re-check, review M2)
U3	C138714	USON-10	Bottom	90	90.0	0.0	1	down-left (-0.41, +1.00)	down-right (+0.41, +1.00)	CHECK in preview
U4	C882825	SOT-23-5	Bottom	0	270.0	270.0	1	down-right (+0.95, +1.14)	down-left (-0.95, +1.14)	CHECK in preview (seen right on 2026-11-07; re-check, review M2)
U6	C201698	MSOP-8	Bottom	90	90.0	0.0	1	down-left (-2.11, +0.98)	down-right (+2.11, +0.98)	CHECK in preview
U7	C3008606	BGA-64_6x6	Bottom	180	180.0	0.0	A1	down-left (-2.27, +2.27)	down-right (+2.27, +2.27)	CHECK in preview; moved -2.5 mm in x in r7d
U8	C138714	USON-10	Bottom	180	180.0	0.0	1	up-left (-1.00, -0.41)	up-right (+1.00, -0.41)	CHECK in preview
U9	C138714	USON-10	Bottom	180	180.0	0.0	1	up-left (-1.00, -0.41)	up-right (+1.00, -0.41)	CHECK in preview
Y1	C725995	OSC3225	Bottom	0	0.0	0.0	1	down-right (+1.05, +0.83)	down-left (-1.05, +0.83)	CHECK in preview
C1	C19702	C0603	Top	90	90.0	0.0	1	down (+0.00, +0.78)		non-polar, either way works
C10	C19702	C0603	Bottom	0	0.0	0.0	1	right (+0.78, +0.00)	left (-0.78, +0.00)	non-polar, either way works
C11	C19702	C0603	Bottom	0	0.0	0.0	1	right (+0.78, +0.00)	left (-0.78, +0.00)	non-polar, either way works
C12	C19702	C0603	Bottom	0	0.0	0.0	1	right (+0.78, +0.00)	left (-0.78, +0.00)	non-polar, either way works

Ref	LCSC	Land	Side	KiCad deg	CPL deg	CPL-KiCad	Pin	Pin 1 from centre, top view (mm)	Same pin in JLC bottom view	Status
C13	C1537	R0402	Bottom	0	0.0	0.0	1	right (+0.48, +0.00)	left (-0.48, +0.00)	non-polar, either way works; new place and rotation in r7d
C14	C19666	C0603	Bottom	90	90.0	0.0	1	down (+0.00, +0.78)	down (-0.00, +0.78)	non-polar, either way works
C15	C1537	R0402	Bottom	0	0.0	0.0	1	right (+0.48, +0.00)	left (-0.48, +0.00)	non-polar, either way works; new place and rotation in r7d
C16	C19666	C0603	Bottom	90	90.0	0.0	1	down (+0.00, +0.78)	down (-0.00, +0.78)	non-polar, either way works; new place and rotation in r7d
C17	C1537	R0402	Bottom	180	180.0	0.0	1	left (-0.48, +0.00)	right (+0.48, +0.00)	non-polar, either way works; new place and rotation in r7d
C18	C19666	C0603	Bottom	0	0.0	0.0	1	right (+0.78, +0.00)	left (-0.78, +0.00)	non-polar, either way works; moved -2.5 mm in x in r7d
C19	C1537	R0402	Bottom	0	0.0	0.0	1	right (+0.48, +0.00)	left (-0.48, +0.00)	non-polar, either way works
C2	C19702	C0603	Top	90	90.0	0.0	1	down (+0.00, +0.78)		non-polar, either way works
C20	C19666	C0603	Bottom	0	0.0	0.0	1	right (+0.78, +0.00)	left (-0.78, +0.00)	non-polar, either way works; moved 0.05 mm in r7c
C21	C1525	R0402	Bottom	90	90.0	0.0	1	down (+0.00, +0.48)	down (-0.00, +0.48)	non-polar, either way works; new place and rotation in r7d
C22	C1525	R0402	Bottom	0	0.0	0.0	1	right (+0.48, +0.00)	left (-0.48, +0.00)	non-polar, either way works; moved -2.5 mm in x in r7d
C23	C1525	R0402	Bottom	0	0.0	0.0	1	right (+0.48, +0.00)	left (-0.48, +0.00)	non-polar, either way works; moved -2.5 mm in x in r7d
C24	C1525	R0402	Bottom	0	0.0	0.0	1	right (+0.48, +0.00)	left (-0.48, +0.00)	non-polar, either way works; moved 0.05 mm in r7c; moved -2.5 mm in x in r7d
C25	C1525	R0402	Bottom	0	0.0	0.0	1	right (+0.48, +0.00)	left (-0.48, +0.00)	non-polar, either way works; moved -2.5 mm in x in r7d
C26	C1525	R0402	Bottom	0	0.0	0.0	1	right (+0.48, +0.00)	left (-0.48, +0.00)	non-polar, either way works; moved -2.5 mm in x in r7d
C27	C1525	R0402	Bottom	0	0.0	0.0	1	right (+0.48, +0.00)	left (-0.48, +0.00)	non-polar, either way works; moved -2.5 mm in x in r7d

Ref	LCSC	Land	Side	KiCad deg	CPL deg	CPL-KiCad	Pin	Pin 1 from centre, top view (mm)	Same pin in JLC bottom view	Status
C29	C1525	R0402	Bottom	90	90.0	0.0	1	down (+0.00, +0.48)	down (-0.00, +0.48)	non-polar, either way works
C3	C1525	R0402	Top	90	90.0	0.0	1	down (+0.00, +0.48)		non-polar, either way works
C32	C1525	R0402	Bottom	0	0.0	0.0	1	right (+0.48, +0.00)	left (-0.48, +0.00)	non-polar, either way works
C33	C1525	R0402	Bottom	0	0.0	0.0	1	right (+0.48, +0.00)	left (-0.48, +0.00)	non-polar, either way works
C4	C1525	R0402	Top	90	90.0	0.0	1	down (+0.00, +0.48)		non-polar, either way works
C40	C19702	C0603	Bottom	0	0.0	0.0	1	right (+0.78, +0.00)	left (-0.78, +0.00)	non-polar, either way works
C41	C1525	R0402	Bottom	0	0.0	0.0	1	right (+0.48, +0.00)	left (-0.48, +0.00)	non-polar, either way works
C5	C52923	R0402	Top	90	90.0	0.0	1	down (+0.00, +0.48)		non-polar, either way works
C50	C1525	R0402	Top	0	0.0	0.0	1	left (-0.48, +0.00)		non-polar, either way works
C6	C1525	R0402	Top	90	90.0	0.0	1	down (+0.00, +0.48)		non-polar, either way works
C60	C45783	C0805	Bottom	90	90.0	0.0	1	down (+0.00, +0.95)	down (-0.00, +0.95)	non-polar, either way works
C61	C45783	C0805	Bottom	90	90.0	0.0	1	down (+0.00, +0.95)	down (-0.00, +0.95)	non-polar, either way works
C62	C45783	C0805	Bottom	90	90.0	0.0	1	down (+0.00, +0.95)	down (-0.00, +0.95)	non-polar, either way works
C63	C1555	R0402	Bottom	0	0.0	0.0	1	right (+0.48, +0.00)	left (-0.48, +0.00)	non-polar, either way works
C64	C52923	R0402	Bottom	90	90.0	0.0	1	down (+0.00, +0.48)	down (-0.00, +0.48)	non-polar, either way works
C65	C45783	C0805	Top	0	0.0	0.0	1	left (-0.95, +0.00)		non-polar, either way works
C66	C45783	C0805	Top	90	90.0	0.0	1	down (+0.00, +0.95)		non-polar, either way works
C67	C45783	C0805	Top	90	90.0	0.0	1	down (+0.00, +0.95)		non-polar, either way works
C68	C1555	R0402	Top	0	0.0	0.0	1	left (-0.48, +0.00)		non-polar, either way works

Ref	LCSC	Land	Side	KiCad deg	CPL deg	CPL-KiCad	Pin	Pin 1 from centre, top view (mm)	Same pin in JLC bottom view	Status
C70	C385032	R0402	Bottom	90	90.0	0.0	1	down (+0.00, +0.48)	down (-0.00, +0.48)	non-polar, either way works
C71	C1525	R0402	Bottom	0	0.0	0.0	1	right (+0.48, +0.00)	left (-0.48, +0.00)	non-polar, either way works
FB2	C98485	C0805	Bottom	0	0.0	0.0	1	right (+0.95, +0.00)	left (-0.95, +0.00)	non-polar, either way works
FB3	C85840	C0805	Bottom	0	0.0	0.0	1	right (+0.95, +0.00)	left (-0.95, +0.00)	non-polar, either way works
FB4	C98485	C0805	Bottom	0	0.0	0.0	1	right (+0.95, +0.00)	left (-0.95, +0.00)	non-polar, either way works
FB5	C85840	C0805	Bottom	0	0.0	0.0	1	right (+0.95, +0.00)	left (-0.95, +0.00)	non-polar, either way works
FB6	C19330	C0603	Bottom	90	90.0	0.0	1	down (+0.00, +0.78)	down (-0.00, +0.78)	non-polar, either way works
L1	C83423	IND4020	Bottom	0	0.0	0.0	1	right (+1.50, +0.00)	left (-1.50, +0.00)	non-polar, either way works
L2	C83423	IND4020	Top	0	0.0	0.0	1	left (-1.50, +0.00)		non-polar, either way works
R1	C25905	R0402	Top	0	0.0	0.0	1	left (-0.48, +0.00)		non-polar, either way works
R10	C25755	R0402	Top	0	0.0	0.0	1	left (-0.48, +0.00)		non-polar, either way works
R11	C25744	R0402	Top	90	90.0	0.0	1	down (+0.00, +0.48)		non-polar, either way works
R12	C25744	R0402	Top	90	90.0	0.0	1	down (+0.00, +0.48)		non-polar, either way works
R13	C25744	R0402	Top	90	90.0	0.0	1	down (+0.00, +0.48)		non-polar, either way works
R2	C25905	R0402	Top	0	0.0	0.0	1	left (-0.48, +0.00)		non-polar, either way works
R20	C4109	R0402	Bottom	0	0.0	0.0	1	right (+0.48, +0.00)	left (-0.48, +0.00)	non-polar, either way works; moved -2.5 mm in x in r7d
R21	C25744	R0402	Bottom	90	90.0	0.0	1	down (+0.00, +0.48)	down (-0.00, +0.48)	non-polar, either way works
R22	C25879	R0402	Bottom	0	0.0	0.0	1	right (+0.48, +0.00)	left (-0.48, +0.00)	non-polar, either way works

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R23	C25879	R0402	Bottom	0	0.0	0.0	1	right (+0.48, +0.00)	left (-0.48, +0.00)	non-polar, either way works
R24	C25092	R0402	Bottom	0	0.0	0.0	1	right (+0.48, +0.00)	left (-0.48, +0.00)	non-polar, either way works
R25	C25744	R0402	Bottom	0	0.0	0.0	1	right (+0.48, +0.00)	left (-0.48, +0.00)	non-polar, either way works
R26	C25744	R0402	Bottom	0	0.0	0.0	1	right (+0.48, +0.00)	left (-0.48, +0.00)	non-polar, either way works
R27	C25744	R0402	Bottom	0	0.0	0.0	1	right (+0.48, +0.00)	left (-0.48, +0.00)	non-polar, either way works
R28	C25744	R0402	Bottom	0	0.0	0.0	1	right (+0.48, +0.00)	left (-0.48, +0.00)	non-polar, either way works; moved 0.08 mm in r7c
R3	C25905	R0402	Top	0	0.0	0.0	1	left (-0.48, +0.00)		non-polar, either way works
R30	C11702	R0402	Bottom	90	90.0	0.0	1	down (+0.00, +0.48)	down (-0.00, +0.48)	non-polar, either way works
R32	C25741	R0402	Bottom	90	90.0	0.0	1	down (+0.00, +0.48)	down (-0.00, +0.48)	non-polar, either way works
R4	C25905	R0402	Top	0	0.0	0.0	1	left (-0.48, +0.00)		non-polar, either way works
R40	C25744	R0402	Bottom	0	0.0	0.0	1	right (+0.48, +0.00)	left (-0.48, +0.00)	non-polar, either way works
R42	C17168	R0402	Bottom	0	0.0	0.0	1	right (+0.48, +0.00)	left (-0.48, +0.00)	non-polar, either way works
R43	C25794	R0402	Bottom	0	0.0	0.0	1	right (+0.48, +0.00)	left (-0.48, +0.00)	non-polar, either way works
R44	C25794	R0402	Bottom	0	0.0	0.0	1	right (+0.48, +0.00)	left (-0.48, +0.00)	non-polar, either way works
R45	C25794	R0402	Bottom	0	0.0	0.0	1	right (+0.48, +0.00)	left (-0.48, +0.00)	non-polar, either way works
R46	C25794	R0402	Bottom	0	0.0	0.0	1	right (+0.48, +0.00)	left (-0.48, +0.00)	non-polar, either way works
R47	C25794	R0402	Bottom	0	0.0	0.0	1	right (+0.48, +0.00)	left (-0.48, +0.00)	non-polar, either way works

Ref	LCSC	Land	Side	KiCad deg	CPL deg	CPL-KiCad	Pin	Pin 1 from centre, top view (mm)	Same pin in JLC bottom view	Status
R50	C25755	R0402	Bottom	90	90.0	0.0	1	down (+0.00, +0.48)	down (-0.00, +0.48)	non-polar, either way works
R51	C122548	R0402	Bottom	90	90.0	0.0	1	down (+0.00, +0.48)	down (-0.00, +0.48)	non-polar, either way works
R52	C25117	R0402	Bottom	0	0.0	0.0	1	right (+0.48, +0.00)	left (-0.48, +0.00)	non-polar, either way works
R53	C25117	R0402	Bottom	0	0.0	0.0	1	right (+0.48, +0.00)	left (-0.48, +0.00)	non-polar, either way works
R54	C25099	R0402	Bottom	0	0.0	0.0	1	right (+0.48, +0.00)	left (-0.48, +0.00)	non-polar, either way works
R55	C25744	R0402	Bottom	0	0.0	0.0	1	right (+0.48, +0.00)	left (-0.48, +0.00)	non-polar, either way works
R60	C26083	R0402	Bottom	0	0.0	0.0	1	right (+0.48, +0.00)	left (-0.48, +0.00)	non-polar, either way works
R61	C728559	R0402	Top	0	0.0	0.0	1	left (-0.48, +0.00)		non-polar, either way works
R62	C319512	R0402	Top	0	0.0	0.0	1	left (-0.48, +0.00)		non-polar, either way works
R63	C728559	R0402	Top	0	0.0	0.0	1	left (-0.48, +0.00)		non-polar, either way works
R65	C17168	R0402	Bottom	90	90.0	0.0	1	down (+0.00, +0.48)	down (-0.00, +0.48)	non-polar, either way works
R66	C25076	R0402	Top	0	0.0	0.0	1	left (-0.48, +0.00)		non-polar, either way works
R67	C25794	R0402	Top	0	0.0	0.0	1	left (-0.48, +0.00)		non-polar, either way works
R68	C247444	C0805	Bottom	0	0.0	0.0	1	right (+0.95, +0.00)	left (-0.95, +0.00)	non-polar, either way works; new in r7c
R71	C25741	R0402	Bottom	90	90.0	0.0	1	down (+0.00, +0.48)	down (-0.00, +0.48)	non-polar, either way works
R72	C25792	R0402	Bottom	90	90.0	0.0	1	down (+0.00, +0.48)	down (-0.00, +0.48)	non-polar, either way works
R73	C11702	R0402	Bottom	90	90.0	0.0	1	down (+0.00, +0.48)	down (-0.00, +0.48)	non-polar, either way works

Ref	LCSC	Land	Side	KiCad deg	CPL deg	CPL-KiCad	Pin	Pin 1 from centre, top view (mm)	Same pin in JLC bottom view	Status
R74	C25092	R0402	Bottom	90	90.0	0.0	1	down (+0.00, +0.48)	down (-0.00, +0.48)	non-polar, either way works
R75	C25092	R0402	Bottom	90	90.0	0.0	1	down (+0.00, +0.48)	down (-0.00, +0.48)	non-polar, either way works
R76	C25092	R0402	Bottom	0	0.0	0.0	1	right (+0.48, +0.00)	left (-0.48, +0.00)	non-polar, either way works
R77	C25092	R0402	Bottom	0	0.0	0.0	1	right (+0.48, +0.00)	left (-0.48, +0.00)	non-polar, either way works
R78	C25092	R0402	Bottom	0	0.0	0.0	1	right (+0.48, +0.00)	left (-0.48, +0.00)	non-polar, either way works
R79	C25092	R0402	Top	270	270.0	0.0	1	up (+0.00, -0.48)		non-polar, either way works
R80	C320843	R0201	Top	270	270.0	0.0	1	up (+0.00, -0.27)		non-polar, either way works
R81	C320843	R0201	Top	270	270.0	0.0	1	up (+0.00, -0.27)		non-polar, either way works
R9	C25741	R0402	Top	0	0.0	0.0	1	left (-0.48, +0.00)		non-polar, either way works